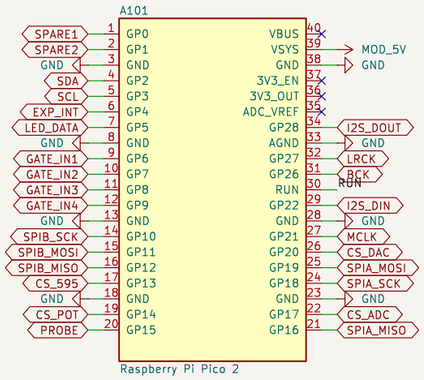
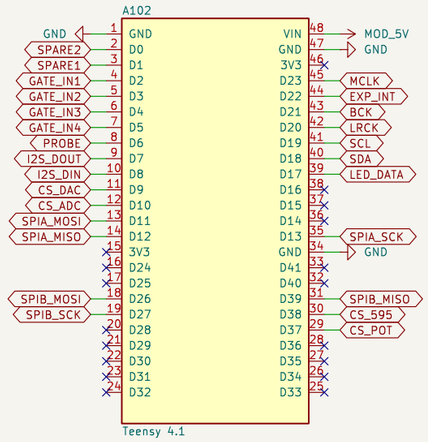
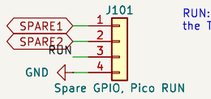
ANALOGUE BOARD REV A — COMPUTE MODULES AND PIN MAP

Both footprints sit on the one board, wired to the same nets; only one module is fitted.
Every signal into a module is 3.3 V: the Teensy 4.1 is not 5 V tolerant.
Pico 2: 5 V reaches VSYS through D203 (Schottky, net MOD_5V), VBUS is left to USB.
Teensy 4.1: cut the VIN-VUSB pads before fitting; VIN takes MOD_5V.
Neither module's own 3.3 V output is used: the board makes its own +3V3A and +3V3D.







RUN: wire a reset button to GND (Pico only; the Teensy has no reset at its edge).

PIN MAP

Net	Function	Pico 2	Teensy 4.1	Why this pin
MCLK	I2S master clock to SCKI1/2	GP21 (pin 27)	23	Pico CLK_GPOUT0 · Teensy MCLK1
BCK	I2S bit clock	GP26 (pin 31)	21	Pico PIO side-set · Teensy BCLK1
LRCK	I2S word clock	GP27 (pin 32)	20	Pico PIO side-set +1 · Teensy LRCLK1
I2S_DOUT	I2S data to codec DIN	GP28 (pin 34)	7	Teensy OUT1A
I2S_DIN	I2S data from codec DOUT	GP22 (pin 29)	8	Teensy IN1
SPIA_SCK	Converter bus clock	GP18 (pin 24)	13	Pico SPI0 SCK · Teensy SCK
SPIA_MOSI	Converter bus data out	GP19 (pin 25)	11	Pico SPI0 TX · Teensy MOSI
SPIA_MISO	Converter bus data in (ADS8688)	GP16 (pin 21)	12	Pico SPI0 RX · Teensy MISO
CS_ADC	ADS8688 chip select	GP17 (pin 22)	10	Pico SPI0 CSn · Teensy CS
CS_DAC	DAC8568 SYNC	GP20 (pin 26)	9	
SPIB_SCK	Panel bus clock	GP10 (pin 14)	27	Pico SPI1 SCK · Teensy SCK1
SPIB_MOSI	Panel bus data out	GP11 (pin 15)	26	Pico SPI1 TX · Teensy MOSI1
SPIB_MISO	Panel bus data in (MCP3208)	GP12 (pin 16)	39	Pico SPI1 RX · Teensy MISO1 (alt pin)
CS_595	74HCT595 latch (RCLK)	GP13 (pin 17)	38	
CS_POT	MCP3208 chip select	GP14 (pin 19)	37	
SDA	I2C data	GP2 (pin 4)	18	Pico I2C1 SDA · Teensy SDA (Wire)
SCL	I2C clock	GP3 (pin 5)	19	Pico I2C1 SCL · Teensy SCL (Wire)
EXP_INT	TCA9555 interrupt, active low	GP4 (pin 6)	22	
GATE_IN1	Gate input 1	GP6 (pin 9)	2	
GATE_IN2	Gate input 2	GP7 (pin 10)	3	
GATE_IN3	Gate input 3	GP8 (pin 11)	4	
GATE_IN4	Gate input 4	GP9 (pin 12)	5	
PROBE	Jack-detect probe	GP15 (pin 20)	6	
LED_DATA	WS2812 data (buffered to 5 V)	GP5 (pin 7)	17	Teensy TX4 (WS2812Serial)
SPARE1	Spare GPIO 1 to wiring field	GP0 (pin 1)	1	Pico UART0 TX · Teensy TX1
SPARE2	Spare GPIO 2 to wiring field	GP1 (pin 2)	0	Pico UART0 RX · Teensy RX1
MOD_5V	Module 5 V in	VSYS (pin 39)	VIN	Through Schottky D203
RUN	Pico reset	RUN (pin 30)	—	To hole J101-3

SHEETS

Power
Power entry, buck, linear rails, test points
File: power.kicad_sch

Codec & audio
PCM3060, two audio ins, two audio outs
File: codec_audio.kicad_sch

CV jacks & ADC
ADS8688, DG412 jack switches, CV holes
File: cv_io.kicad_sch

CV DAC
DAC8568 and the eight ±5 V output stages
File: cv_dac.kicad_sch

Pots & gate outs
MCP3208 pots, 74HCT595 gate outputs
File: pots_gates.kicad_sch

Expander, I2C & LEDs
TCA9555 buttons, EEPROM, Qwiic, WS2812
File: expander_i2c.kicad_sch

Gate ins & jack detect
LM393 gate inputs, probe resistors
File: gate_in.kicad_sch

Edge strips
n8synth edge strips, fiducials
File: edge_strips.kicad_sch

Generated by tools/analogue_board/build.py: edit the generator, not these files
Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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Sheet: /
File: analogue_board.kicad_sch

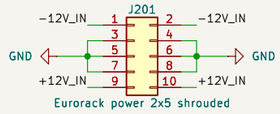
Title: Compute modules and pin map

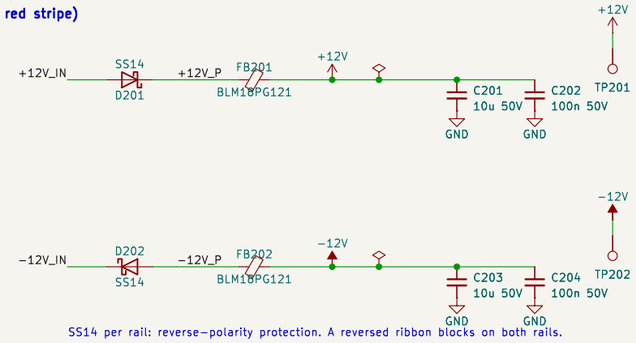
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 1/9

POWER

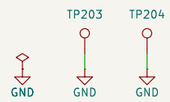
±12 V enter through a 2×5 shrouded header (the bus +5 V is not used). Each rail: series Schottky, ferrite, 10 µF + 100 nF. +5VD comes from a forced-PWM buck (no light-load pulse skipping in the audio band); +5VA from a linear regulator on its own; each 3.3 V rail from its own LDO. A 0 Ω link after every regulator lets a faulty rail be isolated and its current measured.

Eurorack power entry (pin 1 = −12 V, red stripe)





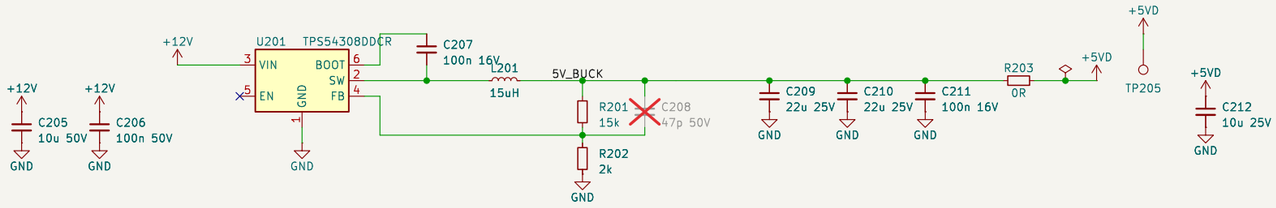
GND test pads



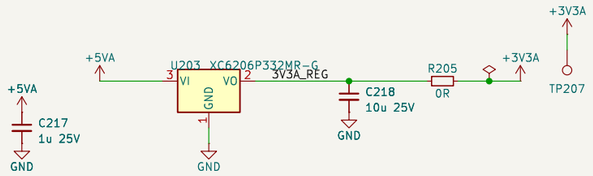
SS14 per rail: reverse-polarity protection. A reversed ribbon blocks on both rails.

+5VD: TPS54308 forced-PWM buck, 350 kHz (TPS54202 is a pin- and value-compatible fallback)

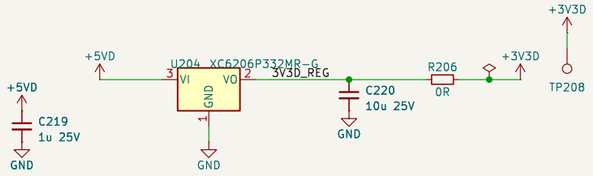
Vout = 0.596 V × (1 + 15k/2k) = 5.07 V. L and Cout from TI's table for 5 V. EN floats (internal pull-up) = on.



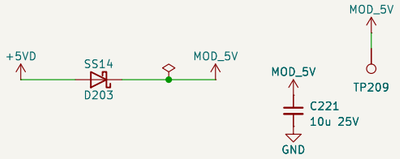
+3V3A: XC6206 from +5VA (DAC8568, MCP3208 and pot reference)



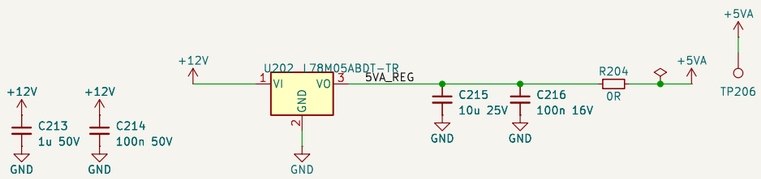
+3V3D: XC6206 from +5VD (codec VDD, ADS8688 DVDD, logic, Qwiic)



MOD_5V: +5VD through a Schottky to Pico VSYS / Teensy VIN



+5VA: L78M05 from +12 V, its own regulator (PCM3060 VCC, ADS8688 AVDD, +3V3A LDO)



Load estimate: +12 V ≈ 120 mA typical, ~350 mA worst case (Teensy at full clock plus a lit LED chain through the buck); −12 V ≈ 25 mA (op-amps). XC6206: 200 mA each; +3V3D ≈ 50 mA, +3V3A ≈ 6 mA.

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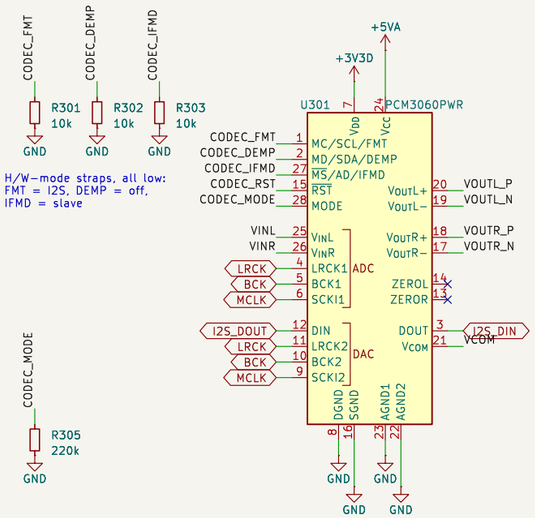
Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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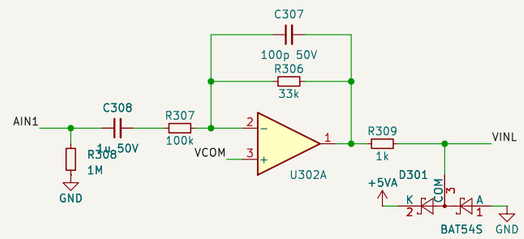
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Title: Power		
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 2/9

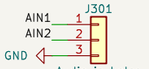
CODEC AND AUDIO

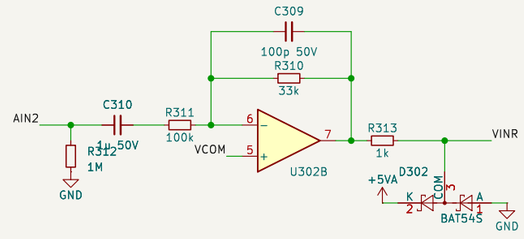
PCM3060 in hardware-control mode: MODE 220 kΩ to DGND = parallel control with differential outputs; FMT low = 24-bit I2S, IFMD low = ADC and DAC both slaves, DEMP low = no de-emphasis. One I2S bus feeds both halves (SCKI1/2, BCK1/2, LRCK1/2 joined). RST is pulled up, so the codec starts by itself 1024 clocks after power-up. ADC full scale 0.6 × VCC = 3 Vpp around VCOM; DAC 8 Vpp differential.



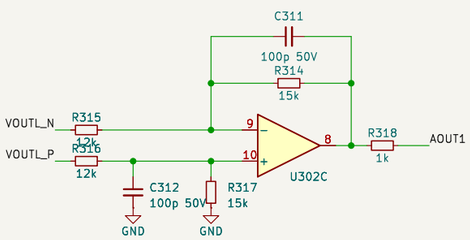
AUDIO INPUTS: 10 Vpp → 3 Vpp. AC-coupled, inverting (firmware flips the sign), ±12 V tolerant. The 1 kΩ into the codec's 10 kΩ input sets part of the gain; BAT54S clamps to the codec rails.

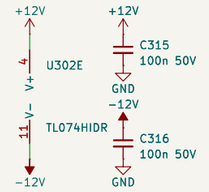




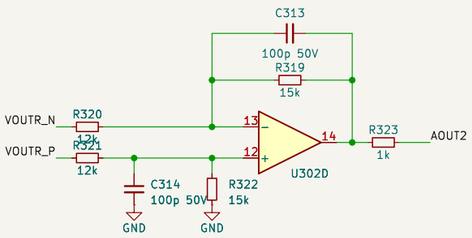


AUDIO OUTPUTS: difference amplifier, gain 15k/12k = 1.25, so 8 Vpp → 10 Vpp (±5 V), DC-coupled. 100 pF COG across each 15 kΩ: a pole at 106 kHz on top of the codec's own RC filter.







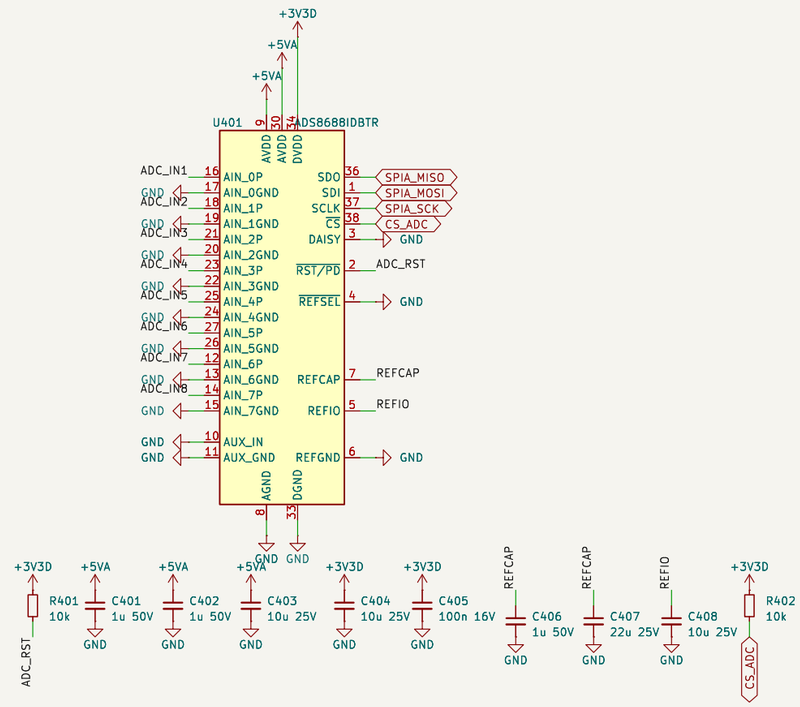


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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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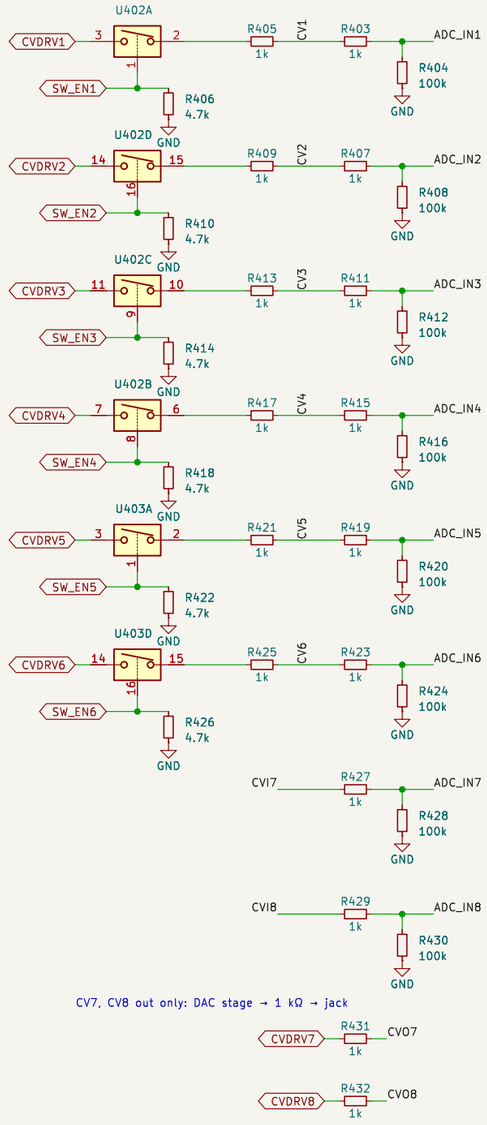
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Title: Codec & audio		
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 3/9

CV JACKS AND ADC

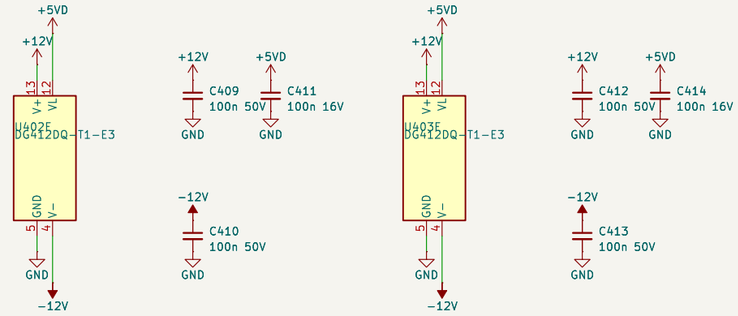
ADS8688: 8 channels, 16-bit, 1 MΩ inputs with ±10.24 / ±5.12 V ranges set per channel in firmware, OVP ±20 V. Every CV jack is read through 1 kΩ; 100 kΩ holds an unpatched input at 0 V. CV1–6 are bidirectional: a DG412 switch (on at logic 1) joins the DAC stage to the jack through another 1 kΩ, so with the switch on the ADC reads the jack itself. The TCA9555 lines have internal pull-ups, so each switch enable has a 4.7 kΩ pull-down that wins (≤ 100 µA × 4.7 kΩ = 0.47 V < 0.8 V): every jack starts as an input. Internal 4.096 V reference.



DAISY, REFSEL, AUX low: stand-alone, internal reference, AUX channel unused.
REFCAP 1 µF + 22 µF and REFIO 10 µF as TI's layout example.



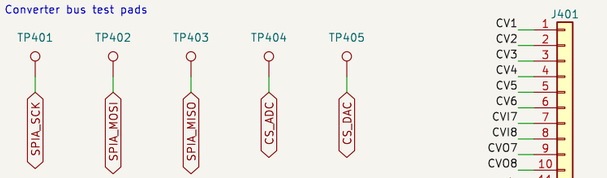
CV7, CV8 out only: DAC stage → 1 kΩ → jack

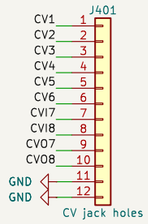


Spare switches (U403 SW3, SW4): off, both sides grounded



Converter bus test pads





CV jack holes

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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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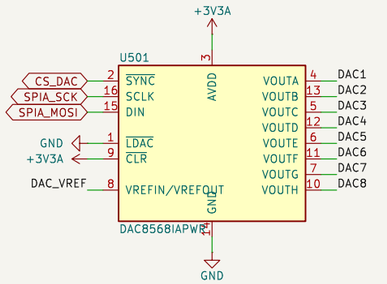
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File: cv_io.kicad_sch

Title: CV jacks & ADC

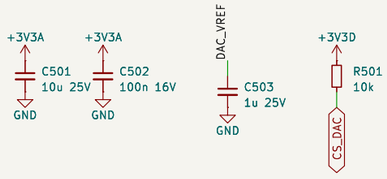
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 4/9

CV DAC

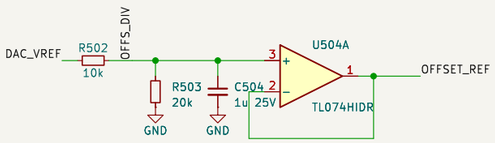
DAC8568 grade A: 0–2.5 V from its internal reference (off at power-up; firmware enables it), outputs reset to zero.
Each stage: non-inverting gain 1 + 30k/10k = 4, offset by OFFSET_REF = 2/3 × VREF through the 10 kΩ:
Vout = 4·Vdac − 3·(2/3)·VREF = −5 V … +5 V. The offset tracks the DAC's own reference, so mid-code is 0 V
whatever the reference does. Before the reference is on, OFFSET_REF is 0 V and every stage sits at 0 V.
±10 V variant (not Rev A): OPA4197 in the same footprint, gain 8, OFFSET_REF = 10/7 V.



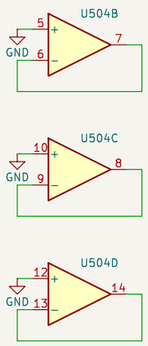
LDAC low: each write updates its output.
CLR high: clear pin unused.

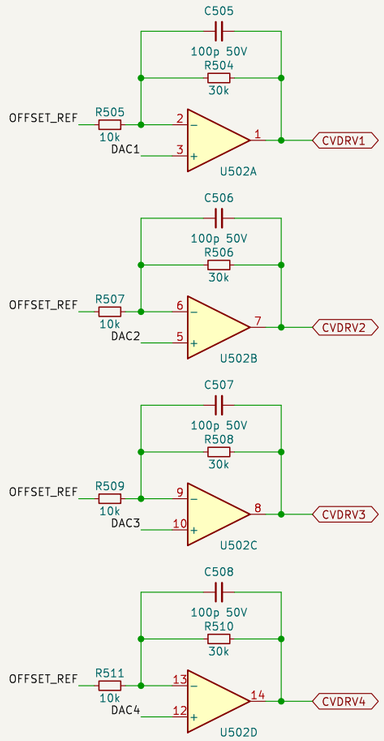


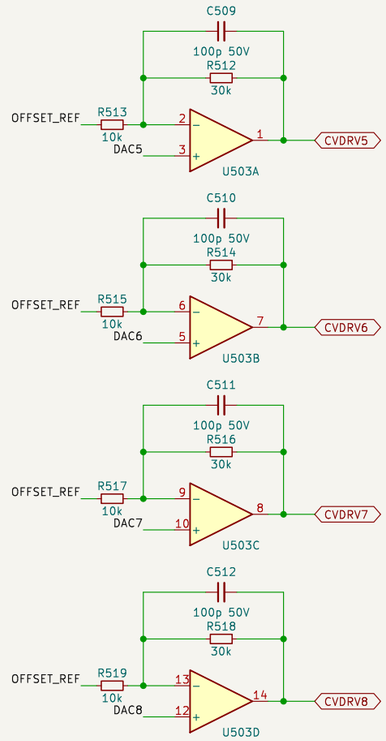
OFFSET_REF = VREF × 20k/(10k + 20k) = 1.667 V, buffered



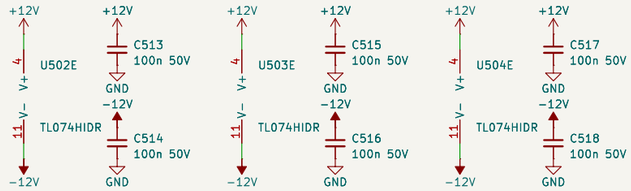
Spare U504 sections: followers with + to GND







CVDRV1–6 go to the DG412 switches; CVDRV7–8 straight to their jacks through 1 kΩ (sheet CV jacks & ADC).

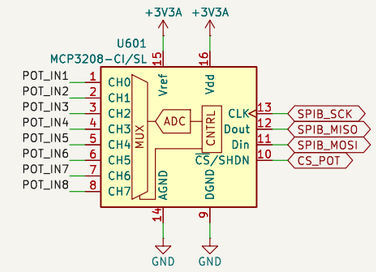


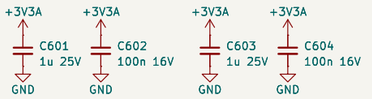
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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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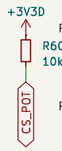
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Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 5/9

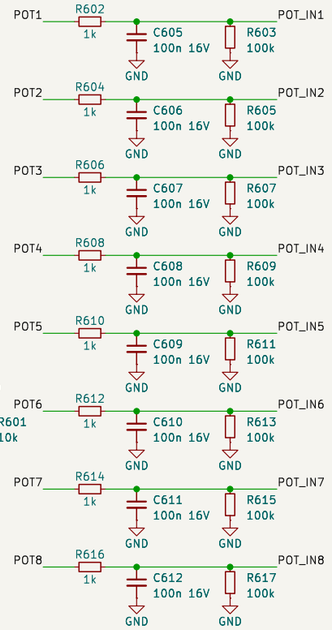
POTS AND GATE OUTPUTS

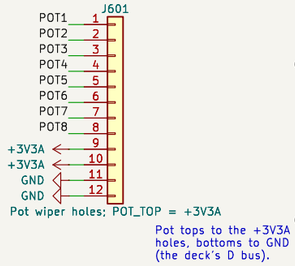
Pots: MCP3208 on +3V3A with VREF = +3V3A, and the pot tops fed from the same rail (POT_TOP holes), so readings are ratiometric. Each wiper: 1 kΩ + 100 nF, 100 kΩ to ground so an unwired channel reads 0.
Gate outs: 74HCT595 at 5 V (VIH 2.0 V, so 3.3 V logic drives it) → 1 kΩ → jack, BAV99 clamps to +5VD and GND.
OE is held high (outputs off) by a pull-up until firmware drives GATE_OE_N low: the 595's register powers up random.

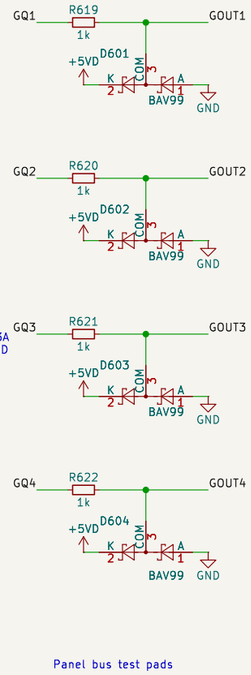


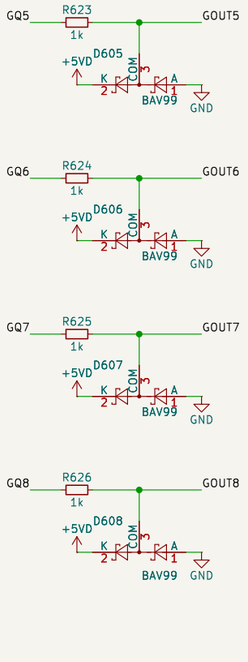


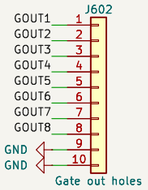




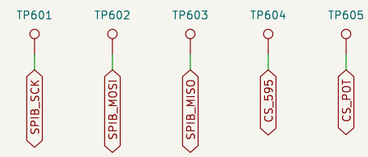


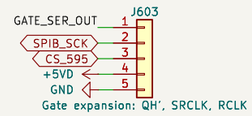




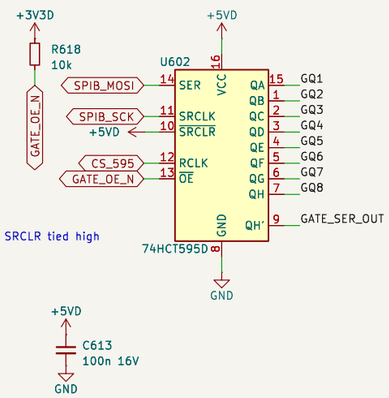


Panel bus test pads





Chain a second 74HCT595 off-board from J603 (Boy's expander needs more gate outs).



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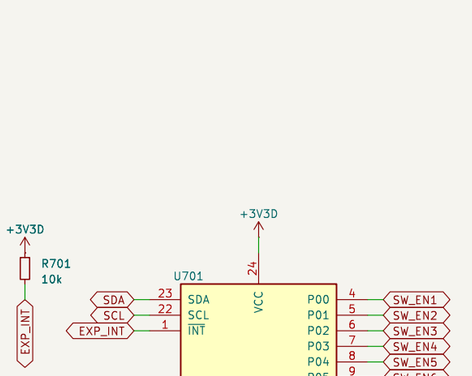
Sheet: /Pots & gate outs/
File: pots_gates.kicad_sch

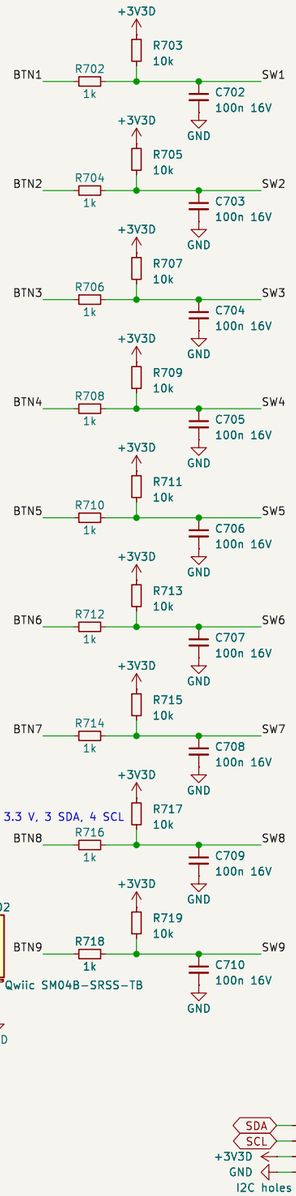
Title: Pots & gate outs

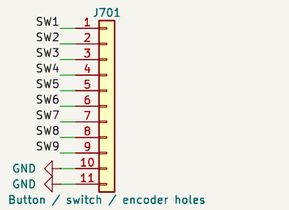
Size: A3	Date: 2026-10-14	**Rev: A**
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EXPANDER, I2C AND LEDS

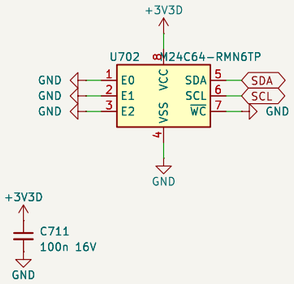
TCA9555 at 0x20, M24C64 at 0x50; a Qwiic display is typically 0x3C/0x3D. 2.2 kΩ I2C pull-ups for 400 kHz.
The TCA9555 powers up with every line an input pulled high internally (~100 kΩ): switch enables and the gate
OE are defined by the pulls on their own sheets. Buttons, switches and an encoder use P07 and P10–P17, each
with EuroPi's 1 kΩ / 10 kΩ / 100 nF (1 ms RC debounce); a button pulls its hole to GND.

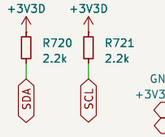




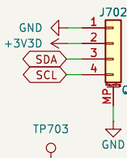


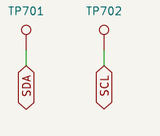
E0–E2 low: 0x50. WC low: writable.

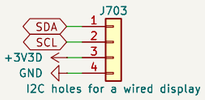




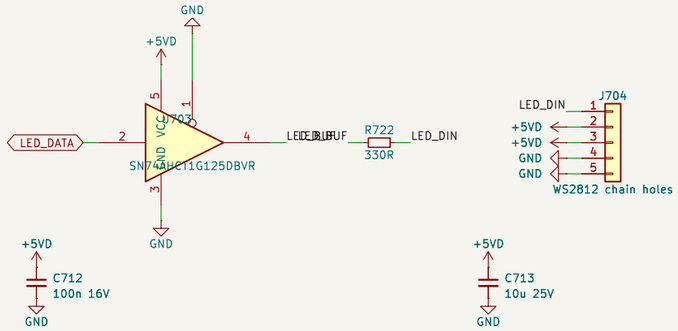
Qwiic: 1 GND, 2 3.3 V, 3 SDA, 4 SCL







WS2812 data: 3.3 V → 74AHCT1G125 at 5 V → 330 Ω → LED_DIN. LED power from the +5VD holes.



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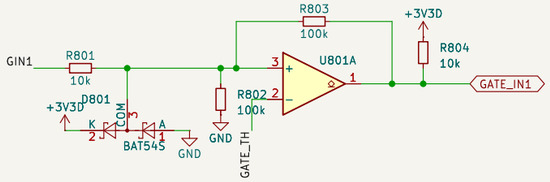
Sheet: /Expander, I2C & LEDs/
File: expander_i2c.kicad_sch

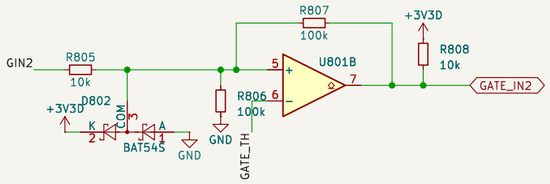
Title: Expander, I2C & LEDs

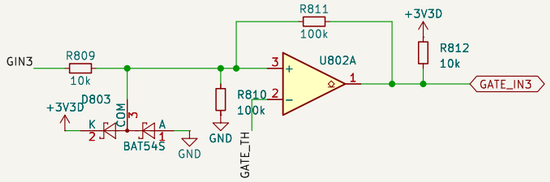
Size: A3	Date: 2026-10-14	**Rev: A**
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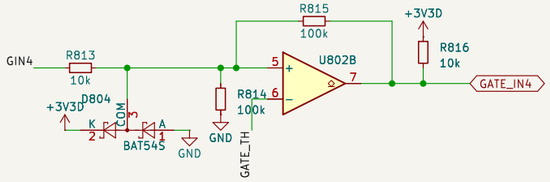
GATE INPUTS AND JACK DETECT

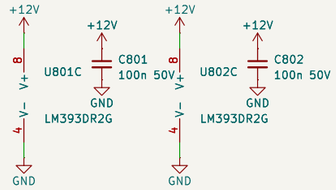
Ogham's clock input: 10 kΩ, BAT54S clamps to +3V3D / GND, 100 kΩ to ground, LM393 against GATE_TH = 1.65 V with 100 kΩ hysteresis. Trips at ≈1.98 V rising, ≈1.67 V falling at the jack. The LM393 runs from +12 V so the clamped input (up to ≈3.6 V) stays inside its common-mode range (VCC − 1.5 V; at 5 V that would be 3.5 V). Open-collector output pulled up to +3V3D: GATE_INn is high while the gate is high, safe for any module.



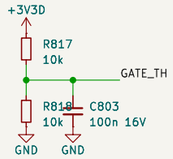






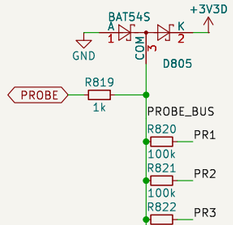


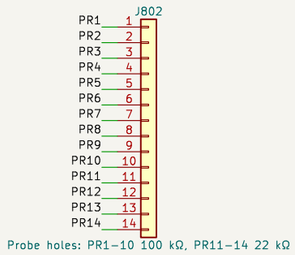
GATE_TH = +3V3D / 2





JACK DETECT (Workshop System Computer): PROBE drives a pseudo-random bit through a resistor to each input's switch (normal) lug. Unpatched, the jack follows the sequence; patched, the lug opens. 100 kΩ for CV and audio inputs; 22 kΩ for gate inputs, whose 1.98 V threshold a 100 kΩ probe cannot reach (it gives ≈1.24 V).





Probe holes: PR1–10 100 kΩ, PR11–14 22 kΩ

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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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Sheet: /Gate ins & jack detect/
File: gate_in.kicad_sch

Title: Gate ins & jack detect

Size: A3	Date: 2026-10-14	**Rev: A**
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EDGE STRIPS

Two n8synth-style edge strips, 40 positions each at 2.54 mm. Each position is an outer hole (the deck's header pin) joined to an inner hole (a wiring point): both pads carry the same pad number, so they share a net. Nothing on the board connects to them: each module is wired by hand from the channel holes to the inner holes. Deck cells occupy positions 1–10, 16–25 and 31–40; 11–15 and 26–30 are free tie points. 10HP: the board sits parallel behind the deck and both strips are used. 6HP: the board sits at 90° and only the left strip meets the deck's right-angle header.

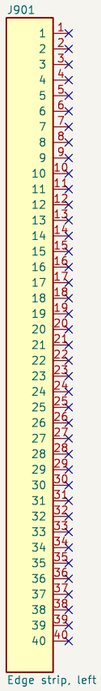
Fiducials for JLCPCB assembly

FID901 Fiducial

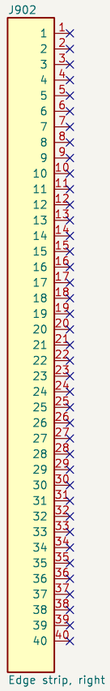
FID902 Fiducial

FID903 Fiducial

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Edge strip, left



Edge strip, right

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Sheet: /Edge strips/
File: edge_strips.kicad_sch

Title: Edge strips

Size: A3	Date: 2026-10-14	**Rev: A**
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